2401CS38

CS3102 - Computer Networks

Lab 1: Simulation of Router Queueing Delay

Input Parameters

1. Packet length (bytes)
2. Source-to-router link bandwidth (bps)
3. Router-to-destination link bandwidth (bps)
4. Source-to-router propagation delay (seconds)
5. Router-to-destination propagation delay (seconds)
6. Router processing delay (seconds per packet)
7. Router queue capacity (number of packets)
8. Packet generation rate, λ (packets/second)
9. Number of packets to simulate
10. Random number seed

Simulation Requirements

- Queueing delay
- Transmission delay
- Propagation delay
- Processing delay
- Total end-to-end delay

Traffic Intensity

The traffic intensity of the output link is given by

$\rho = (\lambda * L) / R$

where

- λ = Packet generation rate (packets/second)
- L = Packet length (bits)
- R = Router-to-destination link bandwidth (bits/second)

For this experiment,

ρ = 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95, 1.0, 1.1, and 1.2

1. Constant Delay Components

The transmission, propagation, and processing delays stay unchanged because they are determined by the nature of the network, such as the speed of hardware devices, the length of cables used, and the size of packets, and hence only queueing delay varies.

2. Delay Spikes Near rho = 1

The router is running at full capacity when the value of traffic intensity approaches 1.0. Due to the nature of network traffic being inherently bursty and random in nature, it tends to cause a backlog of traffic. Since the router is running at full capacity, it cannot afford any idle time to relieve the backlog of traffic.

3. System Overload (rho> 1)

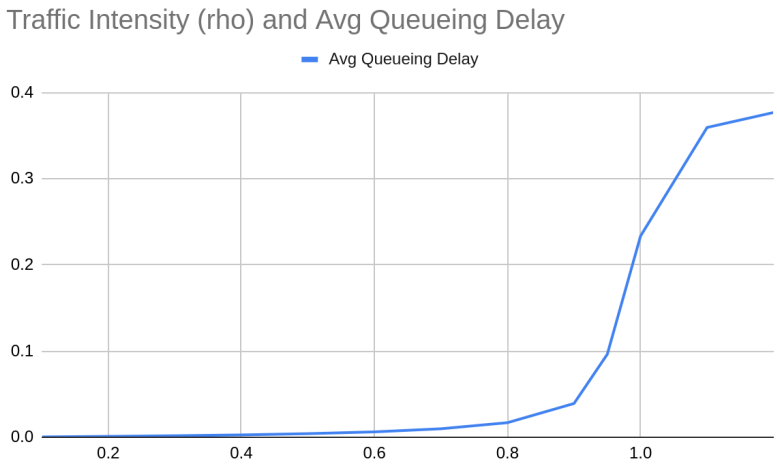
When rho > 1, packets are arriving fundamentally faster than the router can transmit them out.

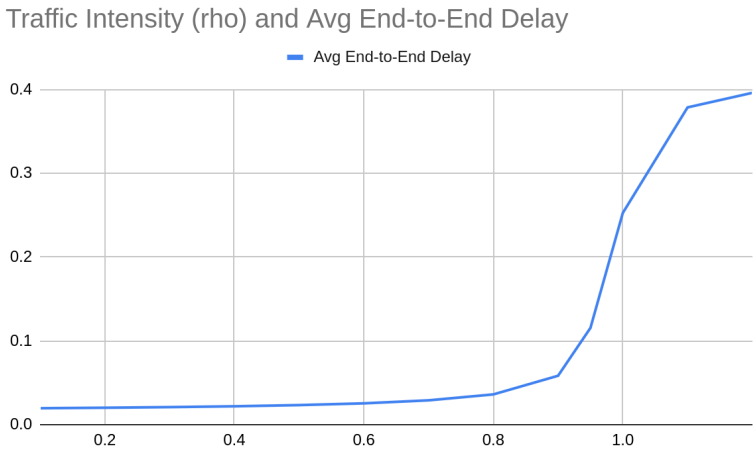
- **Effect on Packet Delay:** Average delay maxes out. The queue stays completely full, so every packet that actually enters the buffer must wait behind a maximum-length line.
- **Effect on Packet Loss:** Packet loss scales up heavily. Because the queue is constantly full, newly arriving packets have nowhere to go and are immediately discarded at the door.

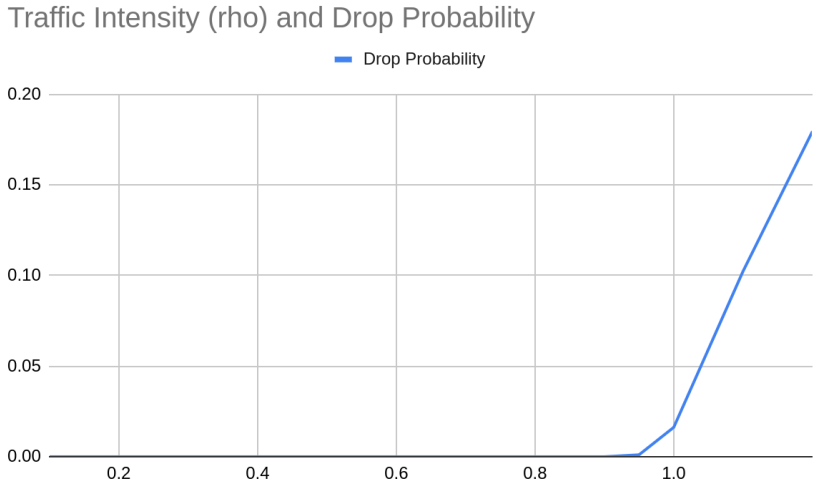
4. Impact of Router Queue Capacity

- **Regarding Average Packet Delay:** Higher capacity means higher delay possibilities. The packets will not be dropped immediately but queued, where there is a big queue in front of them before sending the packets.
- **Regarding Packet Drop Probability:** Higher capacity means lower probabilities of dropping packets. It serves as a buffer that absorbs traffic burst in order not to lose any data.

Results


Traffic Intensity (rho) and Avg Queueing Delay
Avg Queueing Delay
0.4
0.3
0.2
0.1
0.0
0.2
0.4
0.6
0.8
1.0


Traffic Intensity (rho) and Avg End-to-End Delay
Avg End-to-End Delay
0.4
0.3
0.2
0.1
0.0
0.2
0.4
0.6
0.8
1.0


Traffic Intensity (rho) and Drop Probability
Drop Probability
0.20
0.15
0.10
0.05
0.00
0.2
0.4
0.6
0.8
1.0

1. Traffic Intensity (rho) and Average Queueing Delay

- **Observation:** The delay is almost zero and remains fairly constant when the traffic intensity ranges from 0.0 to 0.8. When the intensity reaches 1.0, there is a sudden jump in the mean queuing delay. When the traffic intensity is greater than 1.0, there is an upward trend but the slope becomes flatter.
- **Reason:** At lower arrival rates, the router deals with the packets at a faster rate than they arrive, thus maintaining a small queue size. Once the arrival rate is close to the service rate, there is an accumulation of the packets, and hence the exponential rise in the waiting times. The reason for the small leveling off at 1.0 is that the queue size is finite.

2. Traffic Intensity (rho) and Avg End-to-End Delay

- **Observation:** The graph is an almost exact replica of the Average Queueing Delay graph. The delay is low and constant until rho attains about 0.8 to 0.9 when it spikes up tremendously before settling down a little as rho goes above 1.0.
- **Reason:** Due to the constancy of processing, propagation, and transmission delays, the main variable that contributes to the end-to-end delay is queueing delay. The bottleneck on the queue of the router becomes crucial as the traffic intensity approaches 1.0.

3. Traffic Intensity (rho) and Drop Probability

- **Observation:**Different from the first situation where the packets drop rate was 85%, the drop probability in this case is purely zero up to a traffic intensity of about 0.95. The router does not drop any packets whatsoever before reaching a traffic intensity of 1.0, beyond which the drop rate rises steeply in a linear fashion up to 0.18 (18%).
- **Reason:** The capacity of the router to transmit out is enough to accommodate the incoming packet arrival rate, except where the incoming load exceeds the capacity of the router, which happens when rho is greater than 1.0. At this point, the queue is saturated such that the router starts to drop packets.